Nishan Roy

Software Engineer | AI Engineer | Distributed Systems & Full-Stack
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Education

Narula Institute of Technology 2023 – 2027
B.Tech in Electronics and Communication Engineering *Kolkata, India*

Technical Skills

Languages: Python, TypeScript, JavaScript, Dart, SQL, C, C++
Backend: Node.js, Express.js, FastAPI, Django
Frontend: React, Next.js, Tailwind CSS, Flutter
Databases: PostgreSQL, MongoDB, Redis
AI/ML: TensorFlow, Keras, OpenCV, NumPy, Pandas, LangChain, LangGraph
AI-Assisted Development: Claude Code, OpenAI Codex, Antigravity
DevOps & Automation: Docker, AWS (EC2, S3, Lambda), Git, n8n, Langflow

Experience

Deep Duo Foundation *Kolkata, India*
Software Developer Intern *Feb 2026 – Present*

- Integrated third-party logistics and commerce APIs (Shiprocket, ONDC) into a Node.js backend, handling the full order lifecycle with **idempotent, retry-safe webhook processing** across external platforms.
- Modeled the order flow as an explicit **finite-state machine**, eliminating invalid state transitions and making failure recovery deterministic under partial external outages.

Full-Stack Developer Intern *Jul 2025 – Oct 2025*

- Architected backend services and REST APIs with Node.js, Express.js, and FastAPI for a production web application, applying clean separation of concerns for evolving business logic.
- Modeled PostgreSQL and MongoDB schemas with targeted **compound and covering indexes** for high-read, frequently updated workloads, cutting hot-path query latency.

Projects

Notizz AI | *Deep Duo Foundation Internship – Next.js, Express.js, FastAPI, MongoDB, Flutter, Firebase* **2026**

- Built an AI news-analysis platform, during my **Deep Duo Foundation** internship, flagging bias, detecting near-duplicate coverage, and fact-checking claims across live-aggregated news.
- Shipped end-to-end to production: web app plus a companion **Flutter mobile app**, published and live on the **Google Play Store**.
- Powered by a fine-tuned **DistilBERT** sentiment/bias classifier, **TextRank** summarization, **MinHash + LSH** near-duplicate detection, and **HNSW**-based fact-checking on a dual Express.js/FastAPI backend.

Anvil | *FastAPI, SQLAlchemy, PostgreSQL, Redis, LangChain, Next.js, OpenTelemetry* **2026**

- Built a self-hosted LLM observability and evaluation platform, solving the blind spot most teams ship LLM features with — no visibility into why a chain failed, and no test suite built from real user complaints.
- Delivered the full **trace → complaint → dataset → score → gate** loop as a working, self-hosted product, validated with load tests.
- Runs on FastAPI (async **SQLAlchemy + asyncpg**), **Redis Streams** ingestion decoupled from Postgres, **OpenTelemetry**/Prometheus instrumentation, a **LangChain** multi-provider grading layer, and a Next.js/React dashboard.

ShipIntern | *Next.js, TypeScript, Tailwind CSS, JSearch API* **2025**

- Built an AI-powered internship discovery platform, solving fragmented search across scattered job sources by aggregating listings into one unified, searchable experience.
- Reached **250+ users** during deployment.
- Combines an LLM query-understanding layer that converts natural-language search into structured API parameters, **BM25** relevance ranking, and **Jaro–Winkler** fuzzy matching for cross-source deduplication.

Achievements

- Awarded **Best Intern** out of 10 interns at Deep Duo Foundation for contributions to frontend and backend infrastructure.
- Placed in the **top 20** out of 100+ teams in the NLP Hackathon at Kshitij 2025, IIT Kharagpur.
- Secured **3rd place** out of 50+ participants at Circuitathon 2024, IEM Kolkata.
- Advanced to the finals as a **top 10** finalist at COMSYS Hackathon-4, Jadavpur University.

Certifications & Coursework

- Python Programming – *Infosys*
- API Fundamentals – *Postman*
- AWS Cloud Practitioner Essentials – *AWS*
- Git Fundamentals – *Simplilearn*
- Database Management Systems – *IIT Madras*
- Data Structures & Algorithms – *IIT Madras*